

The IM_MAD section relates to the information monitoring driver for status monitoring. The VM_MAD section deals with the hypervisor driver.

After you have added the worker nodes, your private three-node cloud is ready. However, before you can start launching virtual machines, you need to define a virtual network, and then create and launch a virtual machine.

Virtual networks

You can define custom private networks in OpenNebula, and virtual machines can be associated with several of these virtual networks. Different networks can link specific sets of virtual machines together. You could also permit some machines access to the Internet by connecting them to a specific virtual network.

To define your own virtual network, you need to write a template file. Fortunately, the syntax of a virtual network template file is pretty simple. In OpenNebula, you can define a 'fixed' virtual network or a 'ranged' network. A fixed network specifies certain IP addresses against specific virtual machines identified by their MAC addresses. On the other hand, a ranged network is more like a DHCP configuration, where a base network address is specified and all addresses follow that same pattern. A sample fixed network configuration template is shown below.

```
01 NAME = "Private Cloud"
02 TYPE = FIXED
03 BRIDGE = vbr0
04 LEASES = [IP=192.168.0.1, MAC=50:20:20:20:20]
Ranged Network Template
01 NAME = "wanNetwork"
02 TYPE = RANGED
03 BRIDGE = vbr0
04 NETWORK_SIZE = C
05 NETWORK_ADDRESS = 192.168.0.0
```

In creating a virtual network, I will use the fixed network template file as specified above.

```
01 nebula-user@nebula-cloud-server:~$ onevnet create nebula.
template
02 nebula-user@nebula-cloud-server:~$ onevnet list
03 NID NAME TYPE BRIDGE
04 0 Private Cloud Fixed eth0
```

Creating a KVM virtual machine

To create a KVM virtual machine, you would start by creating the disk file, as follows:

```
dd if=/dev/zero of= ubuntu_server.img bs=1M count=4096
```

Use the *kvm* command to start the installation from the Ubuntu ISO file:

```
kvm -m 512 -cdrom /isos/ubuntu.iso -boot d ubuntu_server.img
```

After you run this command, a QEMU window should pop up and launch the Ubuntu CD boot screen. Select *Install Ubuntu* and follow the instructions.

Submitting the VM to OpenNebula

After the virtual machine is created, the next step is to submit the VM to OpenNebula. Before doing that, you need to carry out a few tasks. To begin with, you must copy the *ubuntu_server.img* created in the last section to the virtual machine repository location. This location is specified in the *oned.conf* file. The variable is called *VM_DIR* and it is commented by default. You'll need to store all images in */opt/vmImages/* on the front-end and update the configuration file accordingly, keeping in mind that you must restart the daemon after every configuration change.

After you've saved the image to the appropriate directory, you can write a VM description for OpenNebula. VM descriptions are mostly hypervisor-specific; however, some general fields are required as shown below:

```
01 NAME = kvmVM #specify the name
02 CPU = 1 # How many CPUs required?
03 MEMORY = 512 # RAM in MB
04 OS = [
05 KERNAL = "/boot/vmlinuz-2.6.32-24-generic", # Kernel
to use
06 INITRD = "/boot/initrd.img-2.6.32-24-generic", #
initrd to use
07 ROOT = "sda1", #Root partition
08 BOOT = hd #boot form harddisk
09 ]
10 DISK = [
11 SOURCE = "/opt/vmImages/ubuntu_server.img",
#location of source image
12 TARGET = "sda1", # mount as partition ]
13 DISK = [
14 TYPE = "swap", #swap drive
15 SIZE = 1024, # size of swap drive
16 TARGET = "sdb" #mount to partition
17 ]
18 NIC = [ NETWORK = "Private Cloud" ] #connect to
specified virtual network, several NIC can be specified, is you
want to connect to several VNs
```

Save the VM description file to the *kvmVM.template*. Note that if you forget to remove the comments, which have been inserted to explain the contents of the template, it will produce parsing errors.

To submit this virtual machine, use the *openvm* command, as shown below:

```
onevm [options] subcommand [parameters]
```

The sub-command arguments associated with *openvm*

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